



Oral-B's AI toothbrush

Oral-B's new toothbrush is capable of rating your brushing habits using Artificial Intelligence. <https://digit.in/jun22-89>



Resident Evil 4 remake

Capcom has announced the remake of Resident Evil 4 is coming to consoles and PC in March 2023. <https://digit.in/jun22-90>

WD Black SN770 NVMe SSD

DRAM-less but powerful

W

D has always maintained a simple colour-coded tiering system with their Red,

Purple, Green, Blue and Black series for the consumer market. For SSDs, it has been the latter three colours. Every year or so, we'd see a new SSD launch happening for each of the series bringing on significant upgrades in terms of transfer speed. This year, we're seeing WD launch multiple drives within a series. The premium WD Black series which is currently helmed by their flagship SN850 is getting a new member in the form of the SN770. It's rated to provide 5.15 GBps sequential read performance and 4.9 GBps sequential write performance. Those numbers also make it a PCIe Gen 4.0 device. Let's have a closer look.

BUILD

WD has found great success with their SSDs over the last couple of years thanks to their in-house controller which came along in the SanDisk acquisition. The controller on the WD SN770 is no different but the architecture is DRAM-less. A premium DRAM-less SSD might sound like an antithetical description but that's exactly what the SN770 is. The NAND chips on the SN770 are Kioxia BiCS5 112L TLC, which are very dense. Moreover, there is only one NAND chip on the 1 TB SKU which is hooked up to the controller with four channels.

The controller is placed right next to the M.2 interface much like what most manufacturers do, and to manage the thermals, the NAND is placed far away on the other end of the SSD. There's a huge amount of space left between the two components to easily allow for thermal dissipation. WD has gone with the same approach previously with SSDs such as WD Blue SN550 and the more recently released SN570. There are no heatsink variants otherwise for the WD Black SN770.

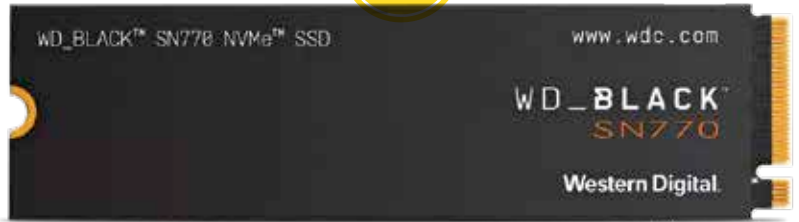
FEATURES

One of the drawbacks of being DRAM-less is that the SSD has to rely on the system memory for most of the caching and for storing a virtual map of your SSD. The DRAM helps your SSD ensure that the SSD NAND undergoes wear evenly. In the absence of a dedicated DRAM, the SSDs allocate a small portion of your system's memory for performing this operation. Most SSDs get up to 64 MB of allocation of the system memory and for caching, they can either use a dynamic cache situated on the SSD NAND itself or use a portion of the system memory again.

However, this approach comes at a cost of increased wear since the NAND is accessed much more frequently in the case of a DRAM-less architecture. This is why warranty terms will be a bit different. In the case of the SN770, the warranty coverage is up to 5 years or 600 TBW (Terabytes Written) for the 1 TB SKU. So if you end up transferring a lot of data, then you will run out sooner than five years. However, most people with SSDs will come nowhere close to hitting 600 TBW in five years.

One thing that the SN770 loses out on is support for 256-bit hardware encryption. Even the SN850 did not have 256-bit AES encryption. Most SSDs, even with encryption, aren't secure thanks to inherent security weaknesses which allow for full data recovery without evening

Price
₹12,399



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knowing the Bitlocker password. So that's there.

PERFORMANCE

Starting off with Crystal Disk Mark, we saw sequential read speeds of 5200+ MBps and write speeds of about 4900+ MBps. Switching to 4K,

we saw read speeds of 80+ MBps and write speeds of 220 MBps. In PCMark, the WD Black SN770 even beat its high-end sibling, the SN850, by a small margin. This was quite unusual considering that the SN770 is a DRAM-less drive. In ATTO, we could see the SN850 easily swing past the SN770. The write speeds were quite similar but the read speeds saw a marked difference.

VERDICT

For a DRAM-less SSD, the WD SN770 sure is a record-breaker of sorts. There might be no heatsink or support for hardware-level encryption but as an additional bucket of storage, the SN770 does make the cut. For an additional INR 2.5K, you can get WD's top of the line SN850 so it's really hard to pick the SN770 and it ideally could have been a little cheaper.

—Mithun Mohandas

SPECIFICATIONS

WD100T3X0E | FORM FACTOR: M.2 2280 | COST/GB: Rs. 12.11 | DIMENSIONS: 80.4 x 23 x 20.4 mm | CONTROLLER: WD Internal | Micron 112-layer TLC BiCS5 NAND | WARRANTY: 5 years

CONTACT

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